

LOW-STYRENE SBR PATENT LANDSCAPE REPORT

1. ABSTRACT

This report synthesizes three primary patents relevant to low-styrene styrene-butadiene rubber (SBR) synthesis and use. The target is defined as SBR embodiments with styrene content below conventional commercial grades, emphasizing reduced aromatic monomer fraction and butadiene-biased copolymer composition. Across the selected patents, two major synthesis families appear: aqueous emulsion polymerization for specialized E-SBR and solution/anionic polymerization for functionalized SBR. EP1967387B1 discloses an aqueous emulsion polymerization prepared E-SBR with 12–16% bound styrene, specifically positioned as a low-styrene elastomer for tire sidewall rubber compositions. US7612144B2 focuses on anionic solution polymerization of styrene/butadiene using sulfur-containing initiators, enabling functional head groups and improved filler interaction; the disclosed SBR examples contain about 21.7–22.2% styrene, which is not low-styrene by the target definition but is still relevant for process and end-group control. US8865829B2 discloses solution polymerization of styrene and butadiene followed by functional termination, with a comparative non-functionalized SBR and functionalized SBRs used in silica/carbon black compounds; the polymerization is conducted in hydrocarbon solvent at 65°C with conversion above 98%. Overall, the landscape shows that low-styrene SBR can be achieved through controlled aqueous emulsion polymerization with reduced bound styrene, while solution polymerization patents emphasize chain-end functionality, filler affinity, and rolling-resistance improvements rather than low styrene content itself. Formulation trends include use in tire sidewalls or tread compounds, reinforcement with carbon black and/or silica, and downstream mixing/cure protocols designed to translate polymer microstructure into lower hysteresis and improved dynamic performance.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP1967387B1
- Patent Title: Rubber composition and tire sidewall with such a composition comprising ...
- Assignee: The Goodyear Tire & Rubber Company
- Jurisdiction: EP
- Publication Year: 2011

- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a specialized emulsion-polymerization-prepared styrene/butadiene rubber (E-SBR) used in tire sidewall rubber compositions. The evidence states that the E-SBR is made by aqueous emulsion polymerization of 1,3-butadiene using an iron/hydroperoxide redox catalyst system, with controlled styrene/butadiene monomer charge to achieve the desired bound styrene level. - Bound styrene content: 14 percent — Example I - Tg (onset glass transition temperature): -65°C — Example I - Mixing temperature: 160°C — Example I - Mixing time: 4 minutes — Example I - Productive mixing temperature: 115°C — Example I - Productive mixing time: 2 minutes — Example I - Cooling temperature: below 40°C — Example I - NR phr: 10 to 40 phr — Claims - Specialized E-SBR phr: 5 to 35 phr — Claims - Cis 1,4-polybutadiene rubber phr: 50 to 70 phr — Claims - Particulate reinforcing filler phr: 30 to 80 phr — Claims - Bound styrene content: 12 to 16 percent — Claims - Tg range: -60°C to -70°C — Claims - Mooney viscosity: 60 to 100 — Claims - Preferred bound styrene content: 13 to 15 percent — Synthesis passages - Preferred Mooney viscosity: 70 to 90 — Synthesis passages - Medium & Water Role: The core polymerization medium is aqueous emulsion polymerization. Water is part of the polymerization medium for E-SBR preparation; no separate downstream water operation is emphasized in the extracted evidence. - styrene content below conventional SBR grades: 12 to 16 percent bound styrene; preferred 13 to 15 percent

Medium & Water Role - The core polymerization medium is aqueous emulsion polymerization. Water is part of the polymerization medium for E-SBR preparation; no separate downstream water operation is emphasized in the extracted evidence. - Core reaction medium: Aqueous emulsion polymerization medium

styrene content below conventional SBR grades - 12 to 16 percent bound styrene; preferred 13 to 15 percent

Relevant Experimental Evidence - Example I: Demonstrated replacement of part of natural rubber with aqueous emulsion polymerization prepared E-SBR containing 14% bound styrene and Tg of -65°C in a sidewall compound. - Example I: Showed compounding of NR, cis 1,4-polybutadiene, E-SBR, carbon black, oil, wax, antioxidants, zinc oxide, sulfur, and accelerator. - Example I: Established that the specialized E-SBR is intended for tire sidewall formulations with low-temperature Tg matching NR. - Example I: Confirmed the E-SBR is an aqueous emulsion polymerization product made via an iron/hydroperoxide redox catalyst system.

Technical Relevance This patent is directly relevant because it discloses a low-styrene E-SBR embodiment with bound styrene in the 12–16% range, which matches the target definition. It also links the polymer to tire sidewall performance, making it highly pertinent to low-styrene SBR synthesis and application.

Patent 2

- Patent Number: US7612144B2
- Patent Title: Use of sulfur containing initiators for anionic polymerization of monomers
- Assignee: Bridgestone Corporation
- Jurisdiction: US
- Publication Year: 2009
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses anionic solution polymerization of styrene and butadiene using sulfur-containing initiators, including 2-lithio-2-methyl-1,3-dithiane and related lithio thioacetal initiators. The process is carried out in hydrocarbon solvent with polar modifier, and the living polymer is coagulated in isopropanol after polymerization; the main purpose is to introduce functional head groups and improve filler interaction rather than to minimize styrene content. - Reactor volume: 1.75 L — Example No. 2 - Hexane: 1.12 kg — Example No. 2 - Styrene solution: 0.48 kg of 33 wt % styrene in hexane — Example No. 2 - Butadiene solution: 2.89 kg of 22.0 wt % butadiene in hexane — Example No. 2 - Reaction temperature: 24°C — Example No. 2 - Modifier: 0.5 mL of 1.6 M cyclic oligomeric oxolanyl alkane modifier in hexane — Example No. 2 - Initiator: 22.63 mL of 0.234 M 2-lithio-2-methyl-1,3-dithiane in tetrahydrofuran — Example No. 2 - Jacket temperature: 54°C — Example No. 2 - Peak batch temperature: 76.5°C — Example No. 2 - Reaction time after heating: 15 minutes — Example No. 2 - Additional time before removal: 25 minutes — Example No. 2 - Mn: 153 kg/mol — Example No. 2 - Mw: 167 kg/mol — Example No. 2 - Tg: -44.4°C — Example No. 2 - Styrene content: 21.7% — Example No. 2 - Block styrene: 1.3% — Example No. 2 - Vinyl: 32.1% — Example No. 2 - 1,4 butadiene incorporation: 46.2% — Example No. 2 - Reaction temperature: 50°C — Example No. 7 - Reaction time: 4 hours — Example No. 7 - Initiator concentration: 1.6 M butyl lithium in hexane — Example No. 7 - Active lithium compound concentration: 0.234 M — Example No. 1 - Synthesis temperature: -78°C — Example No. 1 - Stirring time: 3 hours — Example No. 1 - Overnight storage temperature: -25°C — Example No. 1 - 2-methyl-1,3-dithiane: 10 mL (83.5 mmol) — Example No. 1 - Butyllithium: 55.83 mL of 1.510 M (84.3 mmol) — Example No. 1 - 1-bromo-3-chloropropane: 8.26 mL (83.5 mmol) — Example No. 1 - THF: 350 mL — Example No. 1 - THF for derivatization: 90 mL — Example No. 1 - Polymerization initiator amount: 0.1 to 100 phm — Synthesis passages - Preferred initiator amount: 0.01 to 1 phm — Synthesis passages - Preferred initiator amount: 0.025 to 0.07 phm — Synthesis passages - Monomer ratio: 95:5 to 50:50 — Synthesis passages - Preferred monomer ratio: 95:5 to 65:35 — Synthesis passages - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - styrene content below conventional SBR grades: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Organic solvent / solution polymerization in hydrocarbon solvent with THF modifier

styrene content below conventional SBR grades - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example No. 1: Prepared 2-lithio-2-methyl-1,3-dithiane and confirmed its structure by GC/MS. - Example No. 2: Produced poly(styrene-co-butadiene) with 21.7% styrene using 2-lithio-2-methyl-1,3-dithiane initiator. - Example No. 3: Showed a similar SBR with 22.2% styrene and slightly higher Mn/Mw using the same initiator. - Example No. 7: Demonstrated control polybutadiene synthesis and subsequent coagulation in isopropanol. - Example No. 12: Showed compounded stocks from functionalized polymers and reported reduced tan delta versus control.

Technical Relevance This patent is relevant because it discloses solution/anionic SBR synthesis and the monomer feed, initiator, and temperature controls that govern SBR microstructure. However, the disclosed styrene levels in the SBR examples are around 21.7–22.2%, so the patent is only indirectly relevant to the low-styrene target.

Patent 3

- Patent Number: US8865829B2
- Patent Title: Functionalized polymer, rubber composition and pneumatic tire
- Assignee: The Goodyear Tire & Rubber Company
- Jurisdiction: US
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution polymerization of styrene and butadiene in a 1 gallon reactor at 65°C using n-butyl lithium and TMEDA, followed by termination with isopropanol or functional terminators. The resulting SBRs are then compounded with silica and/or carbon black to evaluate filler interaction and hysteresis, with the invention focused on functionalized end groups rather than low styrene content. - Reactor size: 1 gallon — Example 5 - Reaction temperature: 65°C — Example 5 - TMEDA: 5.7 mmol — Example 5 - n-butyl lithium: 1.6 mmol — Example 5 - Conversion: above 98% — Example 5 - Mixing time: 4 minutes — Example 6 - Mixing speed: 60 rpm — Example 6 - Starting temperature: 140°C — Example 6 - Compression molding time: 1 minute — Example 6 - Second non-productive mix time: 3 minutes — Example 6 - Productive mix starting temperature: 60°C — Example 6 - Productive mix time: 3 minutes — Example 6 - Silica: 65 phr — Table 2 - Oil: 30 phr — Table 2 - Stearic acid: 2 phr — Table 2 - Zinc oxide: 3.5 phr — Table 2 - Sulfur: 1.7 phr — Table 2 - Sulfenamide: 1.6 phr — Table 2 - Diphenylguanidine: 1.5 phr — Table 2 - Antidegradant: 0.75 phr — Table 2 - Polymerization temperature: 65°C — Example 5 - Initiator amount: 1.6 mmol — Example 5 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - styrene content below conventional SBR grades: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Organic solvent / solution polymerization

styrene content below conventional SBR grades - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 5: Demonstrated solution copolymerization of styrene and butadiene and termination at >98% conversion. - Example 5: Showed functionalized SBRs with different end groups and compared them to non-functionalized SBR. - Example 6: Established a silica-filled mixing protocol and measured Payne effect and tan delta. - Example 7: Showed that trithiocarbonate-functionalized SBR reduced strain dependency relative to control in non-productive mixing. - Example 10: Showed that trithiocarbonate-functionalized SBR lowered Payne effect and tan delta compared with non-functionalized control in productive mixing.

Technical Relevance This patent is relevant to low-styrene SBR only as a solution-polymerization reference, because it discloses SBR synthesis and downstream compounding but not a reduced styrene target. Its main contribution is functional termination for improved filler interaction and rolling resistance.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	styrene content below conventional SBR grades	Key Findings
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EP1967387B1	The Goodyear Tire & Rubber Company	2011	The core polymerization medium is aqueous emulsion polymerization. Water is part of the polymerization medium for E-SBR preparation; no separate downstream water operation is emphasized in the extracted evidence.	12 to 16 percent bound styrene; preferred 13 to 15 percent	Example I: Demonstrated replacement of part of natural rubber with aqueous emulsion polymerization prepared E-SBR contain
US7612144B2	Bridgestone Corporation	2009	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	Example No. 1: Prepared 2-lithio-2-methyl-1,3-dithia

US8865829B2	The Goodyear Tire & Rubber Company	2014	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	Example 5: Demonstrated solution copolymerization of styrene and butadiene and termination at >98% conversion.
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SUPPORTING / RELATED PATENTS (SECONDARY)

Supporting Patent 1

- Patent Number: US11130832B2
- Patent Title: Modified conjugated diene-based polymer and rubber composition including the ...
- Assignee: Lg Chem, Ltd.
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent concerns a modified conjugated diene-based polymer and its rubber composition, which is adjacent to SBR but not clearly the requested SBR base material itself.

Supporting Patent 2

- Patent Number: EP3359396B1
- Patent Title: Diene elastomer having a function in the middle of the chain and rubber ...
- Assignee: Compagnie Générale des Etablissements Michelin
- Jurisdiction: EP
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: This is a diene elastomer functionalization patent based on anionic polymerization and post-modification, which is related to but not clearly about SBR as the central material.

Supporting Patent 3

- Patent Number: US9834619B2
- Patent Title: Modified conjugated diene-based polymer, modified rubber composition containing ...
- Assignee: Lg Chem, Ltd.

- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The patent is about a modified conjugated diene-based polymer made by anionic polymerization and end-group coupling, but the packet does not establish that the central invention is SBR rather than a broader diene polymer platform.

Supporting Patent 4

- Patent Number: US20190112458A1
- Patent Title: Solution Polymerized Styrene-Butadiene Rubber-Polyurethane Elastomer Material ...
- Assignee: Beijing University Of Chemical Technology
- Jurisdiction: US
- Classification: SECONDARY (Synthesis relevant; constraints not confirmed)
- Technical Note: The invention is primarily a polyurethane elastomer material that uses solution-polymerized SBR as a soft segment, so SBR is only a component rather than the central subject.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP1967387B1 is the only patent that directly discloses a low-styrene SBR target, with 12–16% bound styrene and a preferred 13–15% range; the other two patents disclose SBRs around 21.7–22.2% styrene or do not state a low-styrene embodiment.
- EP1967387B1 uses aqueous emulsion polymerization with an iron/hydroperoxide redox catalyst system, whereas US7612144B2 and US8865829B2 use solution/anionic polymerization in hydrocarbon solvent.
- US7612144B2 and US8865829B2 both emphasize chain-end functionality and filler interaction; US7612144B2 uses sulfur-containing anionic initiators, while US8865829B2 uses functional terminators after living anionic polymerization.
- US7612144B2 discloses styrene contents of 21.7–22.2% in the SBR examples, which is substantially higher than the low-styrene range in EP1967387B1.
- US8865829B2 discloses polymerization at 65°C with conversion above 98% and termination with isopropanol or functional terminators; its compounding studies focus on silica/carbon black reinforcement and hysteresis reduction rather than styrene reduction.

4. CONCLUSION

The strongest evidence for low-styrene SBR synthesis is EP1967387B1, which directly discloses an aqueous emulsion polymerization prepared E-SBR with 12–16% bound styrene, preferably

13–15%, and a T_g of -60 to -70°C. The other patents are valuable process references for solution-polymerized SBR and functionalization strategies, but their disclosed SBR examples are not low-styrene embodiments. For low-styrene SBR development, the evidence supports aqueous emulsion polymerization with controlled styrene feed and redox initiation as the most directly relevant route, while solution-polymerization patents suggest complementary approaches for end-group control and filler interaction optimization.

5. REFERENCES

- EP1967387B1 | Rubber composition and tire sidewall with such a composition comprising ... | The Goodyear Tire & Rubber Company | EP | 2011 | <https://patents.google.com/patent/EP1967387B1/en>
- US7612144B2 | Use of sulfur containing initiators for anionic polymerization of monomers | Bridgestone Corporation | US | 2009 | <https://patents.google.com/patent/US7612144B2/en>
- US8865829B2 | Functionalized polymer, rubber composition and pneumatic tire | The Goodyear Tire & Rubber Company | US | 2014 | <https://patents.google.com/patent/US8865829B2/en>